



ACCENTURE INNOVATION CHALLENGE 2026 • OFFICIAL TECHNICAL README

ControlPlane.ai

Model-Agnostic Real-Time AI Oversight, Guardrail & Quality Control Tower

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1. Executive Summary & The Enterprise AI Blind Spot

Deploying a Generative AI model was never the hard part. **Trusting what it says, every single time, is.** Traditional APM tools (Datadog, New Relic) only monitor deterministic metrics like CPU load and HTTP 200 codes. When an LLM outputs a fabricated \$10M financial claim or leaks customer credit cards, conventional alarms report HTTP 200 OK. ControlPlane.ai bridges this blind spot by inspecting content safety, factual grounding, and compute efficiency in real time.

\$67.4B

HALLUCINATION LOSS

97%

SECURITY ACCESS GAP

63%

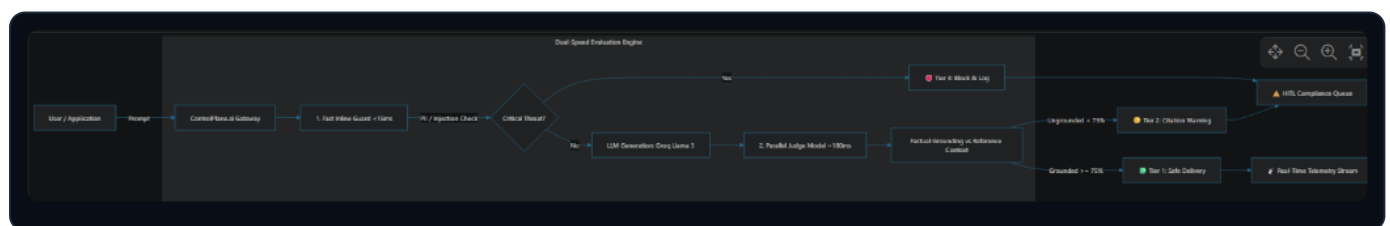
NO QUALITY GATES

<15 ms

FAST GUARD SLA

2. System Architecture: Dual-Speed Evaluation Engine

ControlPlane.ai decouples fast synchronous security checks from deep asynchronous factual reasoning:



3. Mathematical Dynamic Adaptive Sampling

Deep LLM evaluation on 100% of routine traffic is economically unsustainable. ControlPlane.ai dynamically modulates inspection depth based on active anomaly frequency:

DYNAMIC THREAT-TRIGGERED CONTROLLER:

SampleRate(t)

=

25%

Mode)

min(100%, 25% + α · AnomalyRate)

if AnomalyRate_{60s} ≤ 15%

(Nominal

if AnomalyRate_{60s} > 15%

(Surge Mode)

- **Routine Conditions:** Evaluates 25% of baseline traffic, slashing evaluation compute costs by **75%**.
- **Threat Surge / Attack Injections:** Instantly scales up to **85%–100%** inspection depth to shield enterprise infrastructure during attack flurries.

4. Why We Used These Technologies (Design Rationale)

TECHNOLOGY	ARCHITECTURE LAYER	STRATEGIC & ENGINEERING RATIONALE
Groq Cloud (LPU)	Inference Engine	Provides <200ms time-to-first-token inference for 70B/8B models, allowing live parallel Judge verification with zero user lag.
Meta Llama 3 / Compound	Base & Judge Models	Open-weights, enterprise-safe with superior reasoning on structured JSON output and factual comparison benchmarks.
FastAPI (Python 3.10+)	Backend Gateway	Asynchronous ASGI architecture with built-in Pydantic validation, concurrent background workers, and native WebSockets.
React 19 + TypeScript + Vite	Frontend Platform	Strict type safety across the telemetry contract, rapid HMR developer velocity, and optimized component re-rendering.
Tailwind CSS (Dark Obsidian)	Executive UI	High-contrast, accessibility-compliant executive dark aesthetic (#07090E) with light typography and glowing indicators.
Recharts	Visual Analytics	Declarative, GPU-accelerated SVG charting for OWASP threat distribution and latency histograms.

5. OWASP Top 10 for LLMs Security Mapping

OWASP ID	VULNERABILITY NAME	CONTROLPLANE.AI DEFENSIVE MECHANISM
LLM01	Prompt Injection	Synchronous heuristic and delimiter analysis stops jailbreak attempts in <15ms.
LLM02	Sensitive Info Disclosure	Regex, Luhn algorithm, and secret scanners strip/block API keys, credit cards, and SSNs.
LLM03	Hallucination / Misinformation	Parallel Judge model scores factual groundedness against verified reference documents.
LLM04	Data & Model Poisoning	HITL audit trail isolates anomalous model outputs and triggers administrative review.

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OWASP ID	VULNERABILITY NAME	CONTROLPLANE.AI DEFENSIVE MECHANISM
LLM06	Excessive Agency	Dynamic Tiered gating prevents unauthorized downstream execution of unverified commands.
LLM10	Unbounded Consumption	Token metering and adaptive sampling depth caps prevent denial-of-wallet compute spikes.

6. Key Modules & Capabilities

- 🏰 **Control Tower:** Real-time throughput metrics, blocked critical intercepts counter, and live OWASP threat distribution.
- 🧪 **Gateway Lab:** Interactive sandbox with 1-click live presets (Credit Card exfiltration, financial hallucination, prompt injection, and grounded RAG).
- ⚠️ **HITL Queue:** 5-point forensic review modal with 1-click remediation (Approve, Reject, or Redirect to Canned Response) and SOC-2 audit exporter.
- ⚙️ **Policy Studio:** Centralized rule configuration with hallucination sensitivity sliders and encrypted API key management.

7. Quick Start & Execution Guide

STEP	ACTION	COMMANDS
1. Clone Repo	Download project source	<code>git clone https://github.com/tanu99C/ControlPlane-.ai.git</code>
2. Setup .env	Copy template & add Groq key	<code>copy backend\.env.example backend\.env</code>
3. Start Backend	FastAPI server (Port 8000)	<code>cd backend && python -m uvicorn app.main:app --reload</code>
4. Start Frontend	React dev server (Port 5173)	<code>cd frontend && npm install && npm run dev</code>

8. Scalability Roadmap

- **Q4 2026:** Multi-hop agentic guardrails (autonomous agent tool-call interception).
- **Q1 2027:** Air-gapped on-premise deployment package with vLLM / Ollama sidecars.
- **Q2 2027:** Automated continuous model fine-tuning feedback loop based on HITL rejection logs.